DeepSense 6G V2V Dataset, Feature & Target Specification Guide

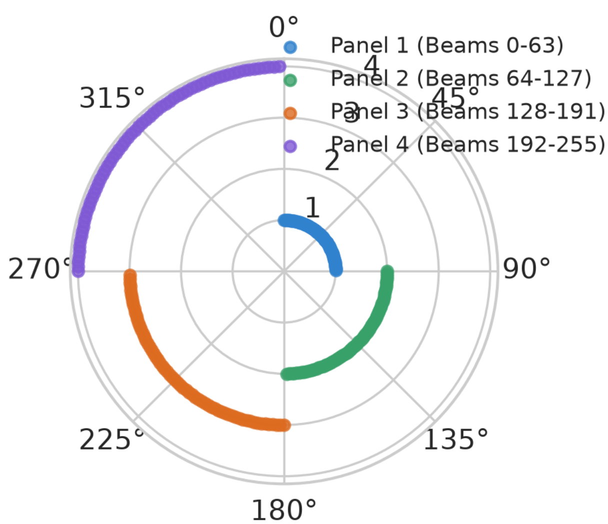
Complete Technical Breakdown of Scenario 36 Modalities, Feature Space, Targets & Visualizations

1. Dataset Overview & Physical Testbed (Scenario 36)

The **DeepSense 6G V2V Dataset** (Scenario 36) was gathered using a real-world multi-sensor vehicle testbed consisting of two mobile units driving in coordinated formations across daytime urban and rural routes at speeds up to 100 km/h.

- **Unit 1 (Receiver Vehicle):** Equipped with a 360° front RGB camera, RTK-GPS, and 4 phased array mmWave receivers operating at 60 GHz.
- **Unit 2 (Transmitter Vehicle):** Equipped with RTK-GPS and a quasi-omnidirectional 60 GHz transmitter.
- **Sample Count:** Scenario 36 contains exactly **24,800 paired time-series samples** recorded at 10 Hz.

DeepSense 256-Beam Codebook Spatial Layout (4 Panels x 64 Beams)

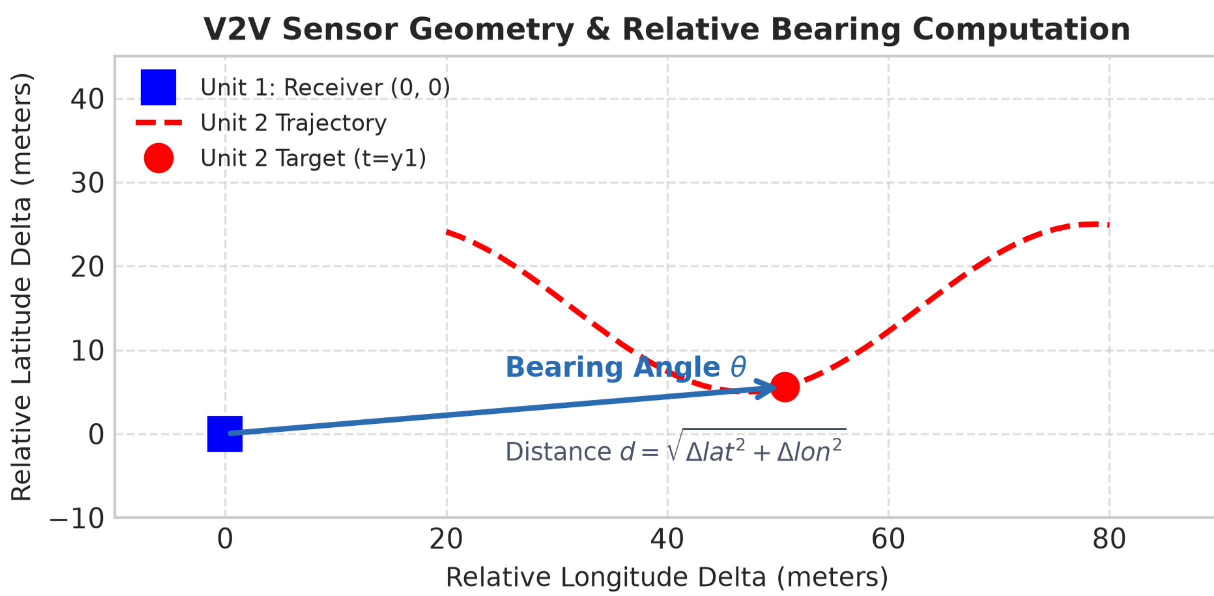


2. Input Feature Space & Feature Engineering Dictionary

The neural network consumes two distinct sensor modalities: **Engineered Spatial GPS Sequences** and **RGB Camera Image Sequences**.

Feature Name	Data Type	Description & Formula	Engineering Rationale
unit1_lat, unit1_lon	Float64	Receiver vehicle RTK-GPS latitude and longitude.	Base spatial positioning.
unit2_lat, unit2_lon	Float64	Target vehicle RTK-GPS latitude and longitude.	Target spatial positioning.
rel_lat, rel_lon	Float32	Δlat = unit2_lat - unit1_lat Δlon = unit2_lon - unit1_lon	Invariance to absolute global location.

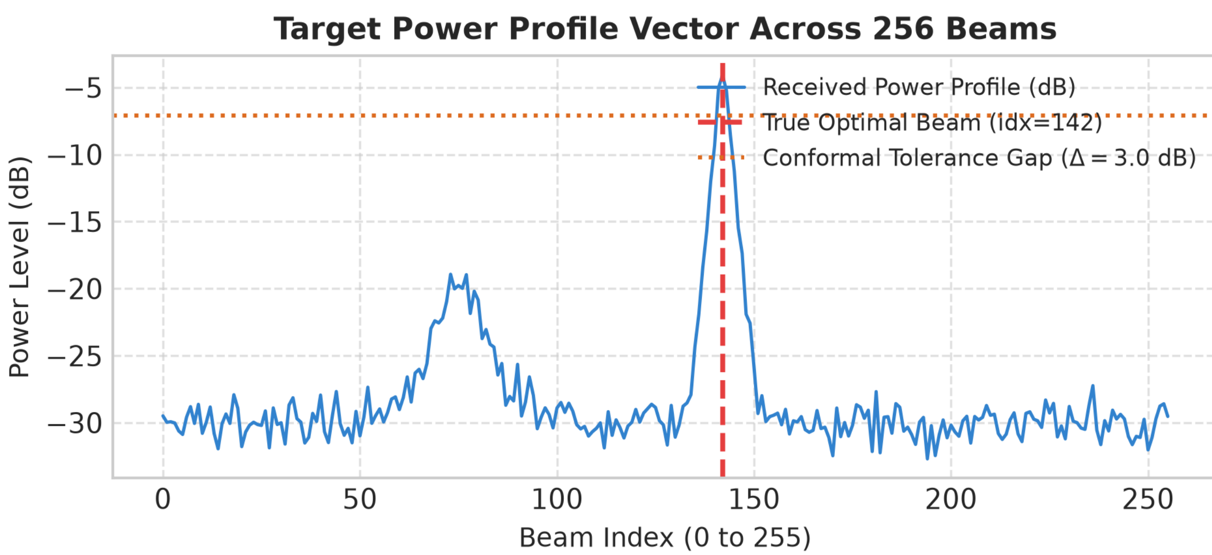
distance	Float32	$d = \sqrt{(\Delta lat^2 + \Delta lon^2)} * 111{,}000$ meters	Encodes path loss & signal attenuation.
bearing	Float32	$\theta = \mathrm{atan2}(\Delta lon, \Delta lat)$ in radians $[-\pi, \pi]$	**Primary spatial predictor of beam angle.**
sin_bearing, cos_bearing	Float32	$\sin(\theta)$, $\cos(\theta)$ continuous bounded encodings	Eliminates $-\pi$ / $+\pi$ boundary discontinuity.
unit1_speed, unit2_speed	Float32	Vehicle ground velocity (m/s)	Encodes rate of angular change.
unit1_heading, unit2_heading	Float32	Vehicle orientation angle relative to North	Aligns body frame to global frame.



3. Target Variables & Ground Truth Labels

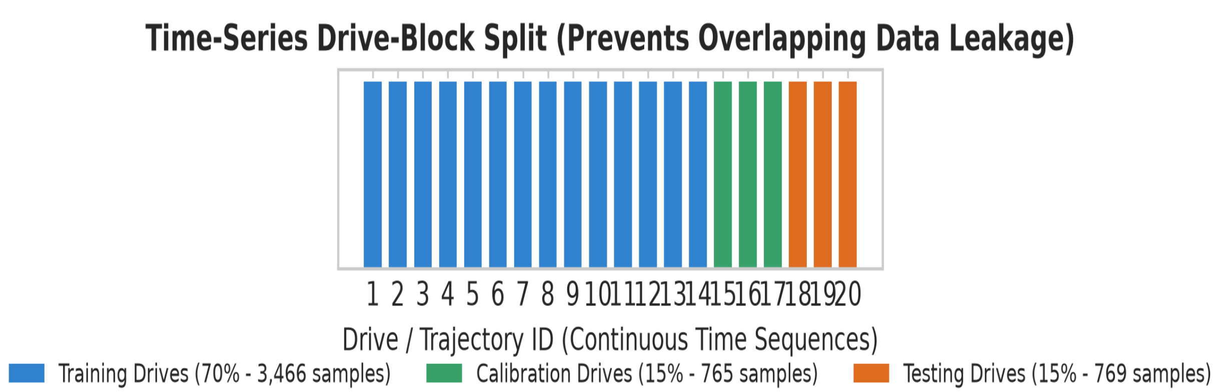
The model is trained as a **Multi-Task Network** that simultaneously outputs discrete classification probabilities and a continuous 256-beam power vector.

Target Variable	Shape & Type	Mathematical Definition	Task Role
true_beam (overall-beam)	Scalar Integer [0, 255]	y_true = argmax_{k} P_k	Primary target for **Cross-Entropy Loss (L_CE)**.
power_profile	Vector (256,) Float32	P = [P_0, P_1, ..., P_{255}] (Linear mW)	Ground truth power across all 256 beams.
true_profile_db	Vector (256,) Float32	P_{dB} = 10 * log10(P_{linear} + 1e-12)	**Target for Mean Squared Error Loss (L_MSE) and Conformal Risk Gap.**



4. Time-Series Drive-Block Dataset Partitioning

Because beam tracking data is a continuous temporal process, standard random k-fold splitting causes severe data leakage across adjacent frames. To guarantee strict evaluation discipline, samples are grouped by **Drive ID** trajectories:



• **Training Drives (70% - 3,466 samples):** Used exclusively for gradient update steps.

- **Calibration Drives (15% - 765 samples):** Held-out to calibrate Conformal Prediction quantile q.
- **Testing Drives (15% - 769 samples):** Held-out for final model evaluation and Conformal PID online validation.